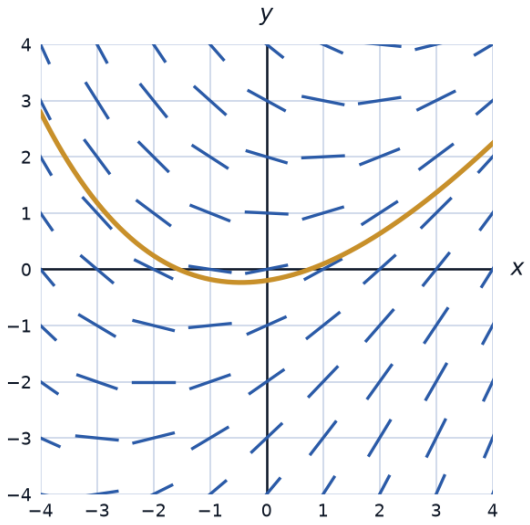
Linear First-Order Equations & the Integrating Factor

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

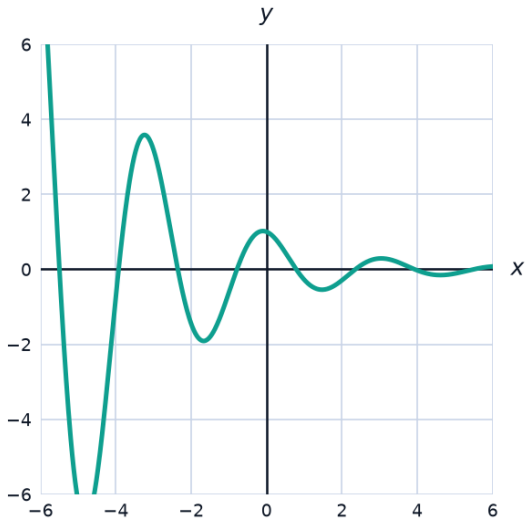
1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the system of linear first-order differential equations given by: $(x'; y') = (1, 2; 3, 0)(x; y)$. Find the eigenvalues of the coefficient matrix.



2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Solve the first-order linear initial value problem $x\frac{dy}{dx} + 3y = \frac{\sin x}{x^2}$ for $x > 0$ with $y(\pi) = 0$. Find $\lim_{x \to 0^+} y(x)$.



3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the first-order linear differential equation $\frac{dy}{dx} + 2y = 4$. If the initial condition is $y(0) = 1$, find the value of $y(1)$.

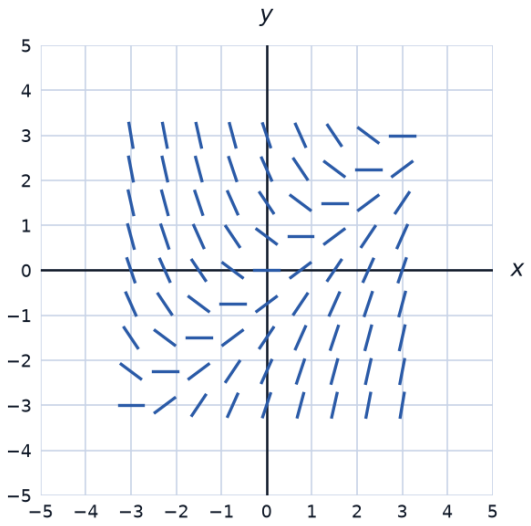
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4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the system of linear differential equations: $\frac{dx}{dt}, = ax + by$; $\frac{dy}{dt}, = cx + dy$ where a, b, c, d are constants. The characteristic equation for the system is $\lambda^2 - (a + d)\lambda + (ad - bc) = 0$. We are given that the eigenvalues of the system are $\lambda_1 = 2$ and $\lambda_2 = -1$. The trace of the coefficient matrix, $T = a + d$, is 1. Determine the value of the determinant of the coefficient matrix, $D = ad - bc$.

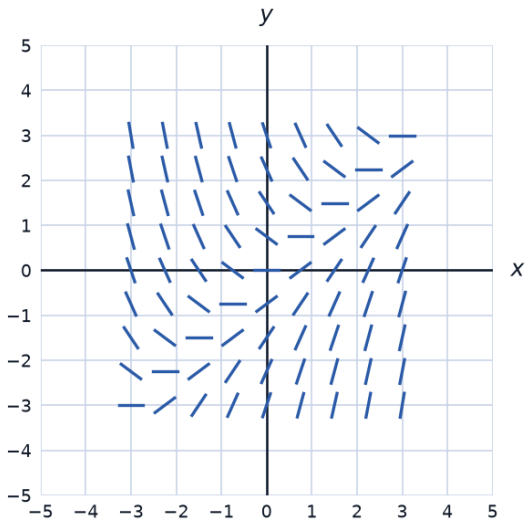
5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find all equilibrium solutions of $\frac{dy}{dt} = y^2 - 4$, classify each as an attractor (sink) or repeller (source), and determine $\lim_{t \to \infty} y(t)$ for the solution with $y(0) = 0$.



6 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

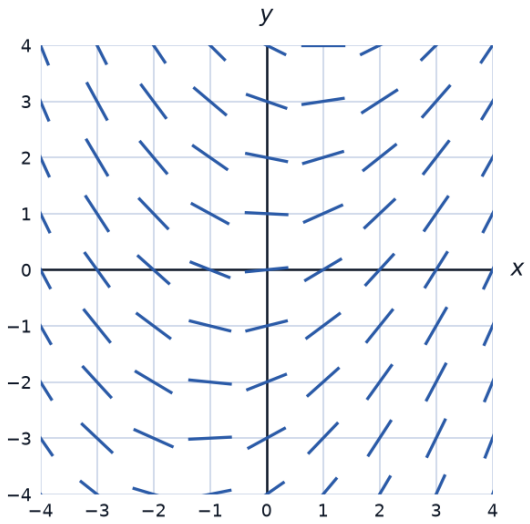
Solve the initial value problem $\frac{dy}{dx} = 2xy^2$ with $y(0) = 1$. Find $y(1/2)$ and state its maximal interval of valid existence.



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7 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Solve the second-order homogeneous initial value problem $y'' - 7y' + 10y = 0$ with $y(0) = 1$ and $y'(0) = -1$. Find $y(1)$.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the differential equation with a regular singular point at $x_0 = 0$: $2x^2y'' + xy' - (x+1)y = 0$, find the indicial equation, determine the two roots $r_1 \geq r_2$, and find the first three non-zero terms of the Frobenius series solution corresponding to r_1.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Solve the defective linear system $\mathbf{x}' = (1, -1; 1, 3)\mathbf{x}$ with $\mathbf{x}(0) = (2; 1)$.

10 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the power series recurrence relation about $x_0 = 0$ for Hermite's differential equation with parameter 2: $y'' - 2xy' + 4y = 0, \quad y(0) = 1, \quad y'(0) = 0$. Prove that the solution terminates to a quadratic polynomial and state the polynomial.

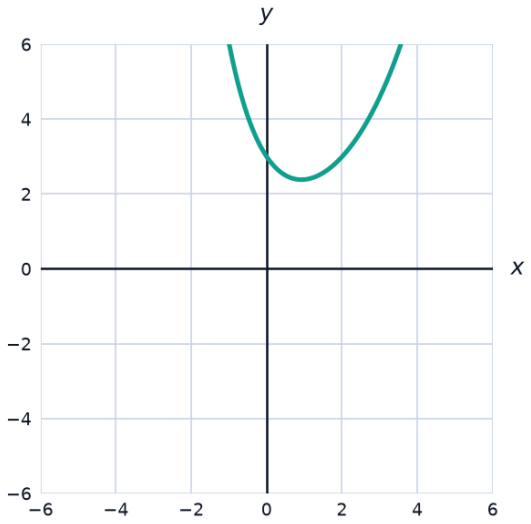
11 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the general solution of the linear system $\mathbf{x}' = (1, 3; 3, 1)\mathbf{x}$. Classify the equilibrium at the origin and determine the cartesian equation of the stable separatrix line.

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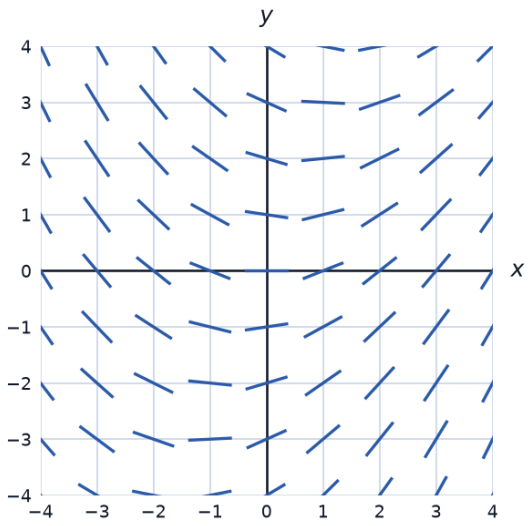
12 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Solve the nonlinear Bernoulli differential equation $\frac{dy}{dx} - \frac{1}{x}y = 2x^2y^3$ for $x > 0$ with $y(1) = 1$. Find $y(2)$.



13 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

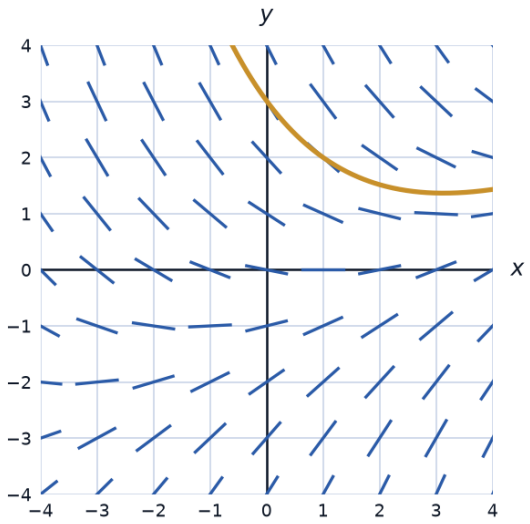
Solve the Cauchy-Euler differential equation $x^2y'' - xy' + 5y = 0$ for $x > 0$ with $y(1) = 2$ and $y'(1) = 4$.



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14 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

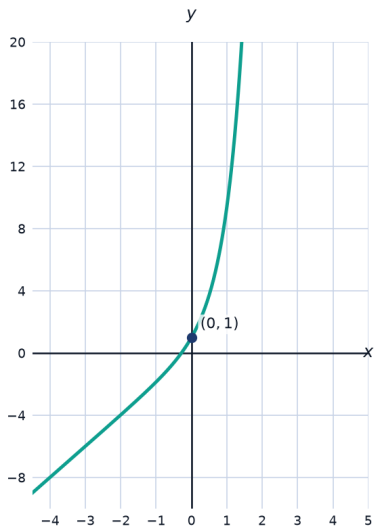
Solve the undamped resonant initial value problem $y'' + 9y = 6\cos(3t)$ with $y(0) = 0$ and $y'(0) = 1$. Identify the secular growth term.



Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Variation of Parameters

Find the general solution of $y'' - 4y' + 3y = 2x + e^{2x}$.



16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Separable Equations

Solve the nonlinear Bernoulli differential equation $\frac{dy}{dx} + \frac{1}{x}y = xy^2$ for $x > 0$ with $y(1) = -1$.

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17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations

Solve the initial value problem $(x^2+1)\frac{dy}{dx}+4xy=x$ with $y(0)=1$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients

[Linear First-Order Equations, Integrating Factors & Nonlinear Reductions · Practice 3] If the governing relation for this system is $g(t)=5t+5$, evaluate $g(8)$ and find the inverse input t when $g(t)=50$.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Variation of Parameters

Solve the Cauchy-Euler differential equation $x^2y''-3xy'+4y=0$ for $x>0$ with initial conditions $y(1)=3$ and $y'(1)=8$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.01 Separable Equations

[Second-Order Linear Homogeneous Equations, Auxiliary Roots & Wronskian Theory · Practice 3] If the governing relation for this system is $g(t)=5t+5$, evaluate $g(8)$ and find the inverse input t when $g(t)=50$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Write the differential equation $x\frac{dy}{dx}+2y=4x^2$ in standard form $\frac{dy}{dx}+P(x)y=Q(x)$ for $x>0$.

a. $\frac{dy}{dx}+\frac{2}{x}y=4x$

b. $\frac{dy}{dx}+2y=4x$

c. $\frac{dy}{dx}+\frac{2}{x}y=4x^2$

d. $\frac{dy}{dx}=4x-2y$

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Compute the integrating factor $I(x)=e^{\int P(x)dx}$ for $\frac{dy}{dx}+\frac{2}{x}y=4x$.

a. x^2

b. $2\ln x$

c. e^{2x}

d. x

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23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Multiply the standard ODE by $I(x) = x^2$ to solve $\frac{d}{dx}[x^2 y] = 4x^3$.

a. $y = x^2 + \frac{C}{x^2}$

b. $y = 4x^2 + C$

c. $y = x^4 + C$

d. $y = x^2 + C$

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the domain of validity for the solution to $x\frac{dy}{dx} + y = 1$ with $y(1) = 2$?

a. $(0, \infty)$

b. $(-\infty, \infty)$

c. $[1, \infty)$

d. $(-\infty, 0)$

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A tank contains 100 L of water and 10 kg of dissolved salt. Pure water flows in at 3 L/min, and the stirred mixture drains at 3 L/min. What is the differential equation for the salt amount $A(t)$ in kg?

a. $\frac{dA}{dt} = -\frac{3}{100}A$

b. $\frac{dA}{dt} = 3 - \frac{3}{100}A$

c. $\frac{dA}{dt} = -3A$

d. $\frac{dA}{dt} = 10 - \frac{3}{100}A$

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the tank above, what is the amount of salt remaining after t minutes?

a. $A(t) = 10e^{-0.03t}$

b. $A(t) = 10 - 0.03t$

c. $A(t) = 10e^{-3t}$

d. $A(t) = 10e^{0.03t}$

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

If salt solution enters at 2 kg/L at a rate of 3 L/min, what is the inflow rate of salt in kg/min?

a. 6 kg/min

b. 2 kg/min

c. 3 kg/min

d. 1.5 kg/min

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In the solution $y(t) = 5 + 4e^{-2t}$, which part represents the transient response and which represents the steady-state?

a. 5 is steady-state, $4e^{-2t}$ is transient

b. $4e^{-2t}$ is steady-state, 5 is transient

c. Both are steady-state

d. The entire response is transient

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the time constant τ for the decaying exponential $e^{-t/\tau} = e^{-5t}$?

a. $\tau = 0.2$ seconds

b. $\tau = 5$ seconds

c. $\tau = 1$ second

d. $\tau = 2.5$ seconds

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

After how many time constants ($t = n\tau$) has a decaying transient decayed to less than 1% of its initial value?

a. 5 time constants ($e^{-5} \approx 0.0067$)

b. 1 time constant

c. 3 time constants

d. 10 time constants

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Consider the first-order linear differential equation $\frac{dy}{dx} + P(x)y = Q(x)$. If the integrating factor for this equation is given by e^{x^2} and $Q(x) = 2xe^{x^2}$, find the value of the constant C in the general solution if $y(0) = 3$. The general solution is of the form $y = e^{-x^2}\left(\int Q(x)e^{-x^2}dx + C\right)$.

a. 1

b. 3

c. 4

d. 6

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

An RC circuit with $R = 4\,\Omega$ and $C = 0.5$ F is connected to $E(t) = 20\cos t$ V. What is the amplitude of the steady-state charge $q_{ss}(t)$?

a. $4\sqrt{5}$ C

b. 5 C

c. $\sqrt{10}$ C

d. $2\sqrt{5}$ C

33 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

An RL circuit has inductance $L = 5$ H and resistance $R = 15\,\Omega$. At $t = 0$, $i(0) = 0$ A. A constant voltage $E = 60$ V is applied. At what time t does the current reach 90% of its steady-state value?

a. $\frac{1}{5}\ln 10$

b. $3\ln 10$

c. $\frac{1}{3}\ln(10/9)$

d. $\frac{1}{3}\ln 10$

34 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

An RL circuit with $L = 1$ H and $R = 2\,\Omega$ has initial current $i(0) = 0$ A. The applied voltage is $E(t) = 10\sin(2t)$ V. What is the amplitude of the steady-state alternating current $i_{ss}(t)$?

a. $5\sqrt{2}$ A

b. $\frac{5}{2}$ A

c. $\frac{5\sqrt{2}}{2}$ A

d. 5 A

35 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

An RL circuit has $L = 3$ H, $R = 9\,\Omega$, and electromotive force $E(t) = 18\cos(3t)$ V. What is the steady-state current $i_{ss}(t)$?

a. $\cos(3t) - \sin(3t)$

b. $\cos(3t) + \sin(3t)$

c. $2\cos(3t) + 2\sin(3t)$

d. $\frac{1}{2}(\cos 3t + \sin 3t)$

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A mixing tank contains 300 liters of brine with 15 kg of salt. Brine containing 0.1 kg/L of salt flows in at 6 L/min, and the mixture drains out at 6 L/min. What is the salt concentration in the tank as $t \to \infty$?

a. 0.05 kg/L

b. 0.1 kg/L

c. 0.2 kg/L

d. 0.15 kg/L

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Consider the second-order linear homogeneous differential equation $y'' - 2y' + 5y = 0$. Let $y_1(x)$ and $y_2(x)$ be two linearly independent solutions. Suppose there exists a solution of the form $y(x) = c_1 y_1(x) + c_2 y_2(x)$ such that $y(0) = 1$ and $y'(0) = 2$. Furthermore, let the Wronskian of y_1 and y_2 be given by $W(y_1, y_2)(x) = e^{2x}$. Determine the value of the expression $y''(0) - 2y'(0) + 5y(0)$.

a. 0

b. e^2

c. 5

d. $2e^2$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients

A 120-gallon tank initially holds 60 gallons of brine containing 30 lb of dissolved salt. Brine containing 1 lb/gal enters at 3 gal/min, while the well-stirred mixture leaves at 1 gal/min. How much salt is in the tank at the exact instant the tank becomes full?

a. 90 lb

b. $120 - 15\sqrt{2}$ lb

c. $120 - 30\sqrt{2}$ lb

d. $120 - 15\sqrt{3}$ lb

39 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Variation of Parameters

A 500-gallon tank initially contains 200 gallons of brine with 40 lb of salt. Brine containing 0.5 lb/gal of salt enters at 6 gal/min, and the well-stirred mixture drains at 4 gal/min. What is the salt concentration in lb/gal when the tank contains 300 gallons?

a. $\frac{37}{45}$ lb/gal

b. $\frac{41}{90}$ lb/gal

c. $\frac{1}{2}$ lb/gal

d. $\frac{37}{90}$ lb/gal

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40 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Separable Equations

A tank contains 100 L of water in which 20 kg of salt is dissolved. Brine containing 0.3 kg/L of salt enters at 5 L/min, and the mixture leaves at 3 L/min. What is the mass of salt in the tank after 50 minutes?

a. $60 - \frac{5\sqrt{2}}{2}$ kg

b. $60 - 5\sqrt{2}$ kg

c. $60 - \frac{5}{2}$ kg

d. $30 - \frac{5\sqrt{2}}{2}$ kg

41 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations

A tank initially holds 100 gal of brine containing 20 lb of salt. Brine containing 1 lb/gal enters at 4 gal/min, and the well-stirred mixture drains out at 6 gal/min. How much salt is in the tank after 25 minutes?

a. 45 lb

b. 40 lb

c. 35 lb

d. 50 lb

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Nonhomogeneous Equations & Undetermined Coefficients

A tank initially contains 100 L of pure water. Brine containing 2 kg/L of salt flows in at 4 L/min, and the thoroughly mixed liquid flows out at 2 L/min. What is the total mass of salt in the tank after 50 minutes?

a. 300 kg

b. 250 kg

c. 350 kg

d. 200 kg

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.01 Variation of Parameters

A tank initially contains 80 L of brine containing 16 kg of salt. Brine containing 0.5 kg/L enters at 6 L/min, and the mixture flows out at 8 L/min. What is the amount of salt in the tank after 20 minutes?

a. $\frac{37}{2}$ kg

b. 19 kg

c. 18 kg

d. $\frac{35}{2}$ kg

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Separable Equations

A 200-gallon tank is half full of distilled water. At $t = 0$, brine containing 3 lb/gal of salt enters at 4 gal/min, while the well-mixed liquid drains out at 2 gal/min. What is the salt concentration in lb/gal at the moment the tank begins to overflow?

a. 2.50 lb/gal

b. 2.00 lb/gal

c. 2.25 lb/gal

d. 1.75 lb/gal

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Second-Order Linear Homogeneous Equations

An RL circuit with inductance $L = 2$ H and resistance $R = 8\,\Omega$ has initial current $i(0) = 0$ A. The applied voltage is $E(t) = 40t$ V. What is the current $i(1)$?

a. $\frac{15}{4} - \frac{5}{4}e^{-4}$ A

b. $5 - \frac{5}{4}e^{-4}$ A

c. $\frac{15}{4} + 5e^{-4}$ A

d. $\frac{15}{4} + \frac{5}{4}e^{-4}$ A

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